

Experiment 2.8: Calculation of Magnetostatic Fields and Energies

Introduction

Numerical integration is often used in cases where analytical solutions are not available. One particular example is the calculation of the magnetic fields from, and the energies of, magnetic structures. In certain cases the fields and energies can be calculated exactly. This is true for systems with simple geometries and uniform magnetization; however, in general it is necessary to use numerical integration. In this experiment you will develop code to calculate the magnetic field and energy of a magnetic system using a finite difference representation.

Magnetostatic field

The magnetostatic field in a vacuum can be calculated using Maxwell's equations for magnetostatics,

$$\nabla \cdot \mathbf{B} = 0 \quad (1)$$

which must be solved subject to the appropriate boundary conditions. If there is a magnetic material present, then we must include the effect of the macroscopic magnetization $\mathbf{M}$ of the material.

If we assume that there is no electrical current then the only boundary condition on $\mathbf{B}$ is that the normal component of the field is continuous if we move between two regions. The presence of a magnetic material adds an additional constraint, that the tangential component of $\mathbf{B} - \mu_0 \mathbf{M}$ is continuous if we move between two regions (e.g. going from vacuum to inside a material).

If the magnetic material has a uniform magnetization, i.e. $\mathbf{M}$ is a uniform vector, then the magnetostatic field at any point outside the magnet is given by

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_S \frac{(\mathbf{r} - \mathbf{r}') \mathbf{M}(\mathbf{r}') \cdot \hat{\mathbf{n}}}{|\mathbf{r} - \mathbf{r}'|^3} dS, \quad (2)$$

where the integral is over the surface S of the magnetic material. Equation 2 gives the magnetic field at any point $\mathbf{r}$, which is outside the magnetic material. If we are inside the magnetic material then the fields are more complex, but we shall not consider that here.

The energy stored in this magnetic field is

$$E_{tot} = \frac{1}{2\mu_0} \int |\mathbf{B}|^2 dV' \quad (3)$$

where the integral is over all space $V' \rightarrow \infty$, excluding the volume of the magnetic material.

Numerical surface & volume integration

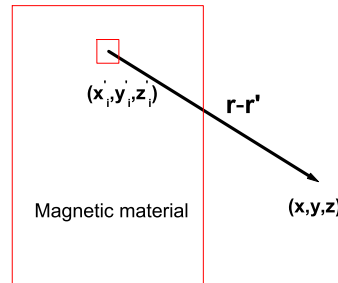


Figure 1: Schematic of the numerical surface integration process

In order to carry out the numerical surface integration we here use a simple approach, outlined in fig 1. Essentially the technique divides the surface of the material into small cells situated at discrete points x'_i, y'_i, z'_i , which represent the coordinates of the centre of the area element. The area is then given by $\Delta S = \hat{\mathbf{n}} \cdot d\mathbf{S}$, so if a particular cell was oriented in the $x - y$ plane at a fixed value of z_i with a normal $\hat{\mathbf{n}}$ in the $+z$ -direction, then $\Delta S = \Delta x \cdot \Delta y$, etc. For cuboid

shapes, with sides that are parallel to the (x, y, z) axes, then there are 6 surfaces to consider with normal vectors $\hat{\mathbf{n}} = \pm\mathbf{k}$ (the sign dependent on whether $\hat{\mathbf{n}}$ points along the positive or negative direction) assuming that the z -direction is normal to the surface, and $\hat{\mathbf{n}} = \pm\mathbf{i}$ and $\hat{\mathbf{n}} = \pm\mathbf{j}$ for the other surfaces.

The function to be integrated is assumed to be piecewise constant over the cell. The surface integral of some function $f(x, y, z)$ can then be approximated using a basic 1st-order Euler scheme:

$$\int_S f(x, y, z) dS = \Delta S \sum_i f(x_i, y_i, z_i), \quad (4)$$

where the summation runs over all the discrete cells on the surface of the integration volume. Note that rather than working with continuous variables we are now using discrete variables, which can be dealt with numerically. For example, the field calculation involves $\mathbf{r} - \mathbf{r}'$. Remembering that $\mathbf{r}'$ is the integration variable over the surface of the magnetic material, we have

$$\mathbf{r} - \mathbf{r}' = (x - x_i)\mathbf{i} + (y - y_i)\mathbf{j} + (z - z_i)\mathbf{k}, \quad (5)$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the usual unit vectors, and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the point at which the field is to be calculated.

In the same way, we can also do volume integrals, summing over volume elements $\Delta V = \Delta x \Delta y \Delta z$ over all points x'_i, y'_i, z'_i within the volume of the material:

$$\int_V f(x, y, z) dV = \Delta V \sum_i f(x_i, y_i, z_i), \quad (6)$$

Objectives

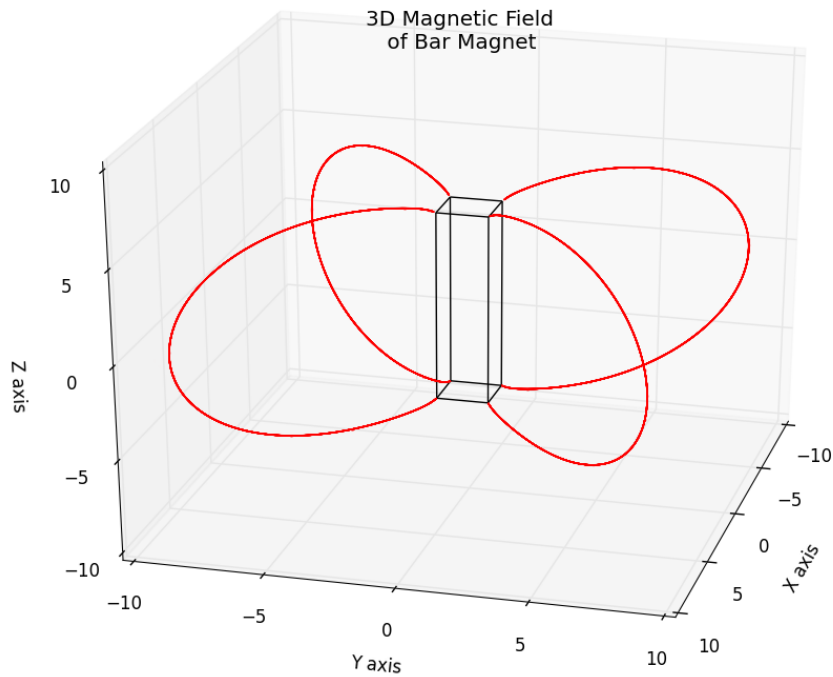
- To develop code for the numerical calculation of the magnetic field of a simple magnet using surface integration
- To visualize a single magnetic field line in 3D
- To develop code for the numerical calculation of the magnetic energy of a simple magnet using volume integration

Experiment

This is a 2-week experiment. The relative amount of time you should spend on each section is shown as [X%].

1. Tracing a field line [75%]:

- (a) Write a program to calculate the surface integral for the magnetic field $\mathbf{B}(\mathbf{r})$ at a point $\mathbf{r} = (x, y, z)$ at any point outside a bar magnet using Eq 2 and summing over all the relevant faces using Eq 4. How can you test that it is correct? A simple test, that will correspond with your expectations, is to plot the path of a single field line through space. For a simple bar magnet, a cuboid of size $2 \times 2 \times 10$ centred on the origin, with uniform magnetization $\mathbf{M} = (0, 0, 1)$, it should look something like this:



You should choose a starting point $\mathbf{r}$ near the upper z -surface and calculate $\mathbf{B}(\mathbf{r})$ at that point. Then move from $\mathbf{r} \rightarrow \mathbf{r} + d\mathbf{r}$ where $d\mathbf{r} \propto \hat{\mathbf{B}}(\mathbf{r})$ and repeat, until the line either goes too far from the material or tries to enter the material. The accuracy of the line tracing will depend upon the size of the step $d\mathbf{r}$ and the quality of each surface integral to get each value of $\mathbf{B}(\mathbf{r})$.

- (b) For the cuboid bar used in Experiment 1a, centred on the origin as in the figure, what is the field $\mathbf{B}(\mathbf{r})$ at a point $\mathbf{r} = (1, 1, 1)$? The accuracy of the field calculation will depend upon the quality of the surface integral, i.e. the number of points in the surfaces you are summing over.

2. Volume integral [25%]

- (a) Now extend your code to calculate the total energy stored in the magnetic field around the same bar magnet using Eq. 3. You will need to be careful about the quality of each surface integral to get each $\mathbf{B}(\mathbf{r})$, **and** the number of points and size of the volume, to get a good answer.

Coding Hints

- code design
 - There is an obvious candidate for code reuse in this experiment – the calculation of the field $\mathbf{B}(\mathbf{r})$ for any arbitrary point $\mathbf{r}$ given a particular size /shape of magnet of given magnetization $\mathbf{M}$. This can then be used for the field line following **and** the volume integration to get the energy. Can you think of any other design ideas?
- visualization
 - The above figure was made using matplotlib, where the data was written out to the file '3D_plot.txt' as rx ry rz Bx By Bz |B| and plotted using the quiver function. See the "3D plotting" section under "Course Documents" on the VLE for more information.
- accuracy
 - The accuracy of the field at any point will depend on the number of points $\mathbf{r}'$ on each surface. The accuracy of the field line tracing and the volume integral will depend on the size of the step $d\mathbf{r}$ used to generate the points $\mathbf{r}$. How can you check that you have a good calculation?
- Compiler flags
 - Finally, don't forget to use appropriate compiler flags as before.